Mini Project 12 — Securities Investor Grievance Routing: Project Plan

UE24AM342AA5 ML System Design & AgentOps · PES1UG24AM244 · PES1UG24AM245 · PES1UG24AM257 · PES1UG24AM273

1. Problem and scope

A multi-agent service that takes a free-text investor complaint, decides which regulator and entity own it, and recommends a routing with circular-level citations. A grievance officer decides; the system never disposes of a complaint. SEBI and RBI frameworks list what they cover, not what they exclude, so investment wording about a bank product (the 4.1 fixed-deposit case) wrongly retrieves SEBI text. The system must escalate when a complaint splits across regulators or the entity is outside the corpus. Out of scope: model training, cloud deployment, real regulator portals.

2. Agent design

Agent	Responsibility	Hand-off
Research agent	Classifies domain; retrieves the apparent provision, checks for modifying provisions, re-queries from what it read	Evidence (passages + queries) to verifier
Verifier agent	Drafts determination; checks every citation resolves to retrieved text; routes or escalates	Routing result, or back to research with a new strategy

Tools: corpus search, clause lookup by ID, request-field lookup (agent-chosen, not a fixed stage). **Limits:** caps on steps, tool calls and wall-clock time. An unverified or incomplete run is escalated, never returned as confident.

3. Architecture

One `docker compose up`; runs are asynchronous (run ID returned, client polls status). Model behind our own interface; no secrets in the repo.

Service	Role	Why separate
API (FastAPI)	Validate, quarantine, status	Stateless, scales independently
Agent worker	Runs LangGraph agents	Scaled with --scale
Redis	Queue, cache, rate limits	Decouples submit from execution
Vector store + index job	Chroma/FAISS, CLI pipeline	Re-embed independently
Prometheus, Grafana, Jaeger	Metrics and traces	Isolated from request path
LLM stub	Fixed reply, realistic delay	Load tests measure our system

4. Corpus and retrieval

Corpus: SEBI investor grievance master circular, SEBI (Mutual Funds) Regulations investor-servicing chapter, RBI Integrated Ombudsman Scheme. LangChain with Chroma/FAISS and local all-MiniLM-L6-v2 embeddings; clause-level chunks with metadata (document, clause ID, regulator, scope). An idempotent CLI pipeline logs build time, versions, chunk count; re-embedding is per document. To counter the cross-framework confusion, the agent searches both frameworks and derives a second query targeting scope and exclusion provisions.

5. The 4.1 case (reference answer)

1. Classify: bank fixed deposit, entity is a bank, not a SEBI intermediary.
2. SEBI text is retrieved first; the agent finds no bank deposit in its covered matters and re-searches ("bank", "deposit").
3. It retrieves the RBI Ombudsman Scheme provisions; the verifier confirms citations.
4. Route: **refer to RBI Ombudsman**, with citations and basis.

6. Evaluation, LLMOps and operations

- **Eval (L2, L3):** 20+ scenarios incl. 4.1, CLI batch; groundedness, context relevance, faithfulness, hallucination rate before and after one change; two prompt versions compared.
- **Safety and cost (L4, L5):** injection defence, output filter, toxicity checks; per-step token cost.
- **Observability (D5):** correlation IDs, OpenTelemetry, Prometheus, Grafana; runs reconstructible by ID.

- **SLO (D6):** p95 declared before measuring; Locust against the stub; report failing concurrency and effect of --scale.
- **Reliability (A5, D7):** timeouts, backoff, circuit breaker, caching, rate limiting.

7. Priorities

Phase	Focus	Requirements
1. Working path	Pipeline, FastAPI + queue, one agent, web client, compose up	D2, D3, A4, A8, F1
2. Correctness	Validation, second agent, verification, escalation, 4.1	D1, A1–A3, F5, F7
3. Trust	Eval set, prompt versions, injection defence	L2, L3, L4
4. Operations	Tracing, Locust, breaker, cache, runbook	D5–D8, A5, A7
5. Stretch	Cost report, feedback, live scoring	L5, L6